15 Total No. of printed pages = 4

**1 MCA 2 CP&PS**

**2011**

(December)

# COMPUTER PROGRAMMING AND PROBLEM SOLVING

Paper : 102

Full Marks – 75

Pass Marks – 34

Time – Three hours

The figures in the margin indicate full marks for the questions.

Answer any *five* questions.

1. (a) What are the characteristics of fourth generation computer ? (P-119) 3

(b) Give answer with True or False : 5

(i) 4 bits equal to 2 Nibble. F

(ii) ASCII is commonly used to translate the object programme to source programme. F



Scanned with CamScanner

(iii) C language is sometimes called middle level language.

(iv) High level languages are also termed as third generation language.

(v) Operating system is an interface between the user and the system.

(c) What do you mean by flowchart and algorithm? Develop flowchart and algorithm for finding factorial of an integer. 2+2+3=7

2. (a) What are the differences between RAM and ROM ? 4

(b) What do you mean by system software and application software ? What is an operating system ? Mention its function. 7

(c) What do you mean by memory hierarchy ? Describe cache memory. 4

3. (a) How many types of control / conditional statement available in C language ? Describe each with a suitable example. 6

(b) What is the difference between break and continue statement ? 3



(c) What do you mean by static and dynamic memory allocation ? What are the function's use for memory allocation dynamically in C language ? 6

(a) Write a C programme to convert a binary number into a decimal number. 6

(b) What is a structure ? How does a structure differ from an array ? 4

(c) What do you mean by local variable, global variable, formal argument and actual argument ? 5

(a) What is meant by recursion ? Write a recursion function to display the sum of the digits of a number. 7

(b) Explain the four string handling function in C. 4

(c) Explain how a file is opened and closed. 4

(a) What do you mean by sorting and searching? Write a C programme to sort 10 elements with bubble sort technique. 5

(b) Write a C programme to create a text file "CPPS.TXT", and input 'C is a middle level language'. 5



Scanned with CamScanner

(c) What do you mean by void function ?

(d) Differentiate built - in function and u define function.

7. Write short answer (any *three*) : 5×3=

(i) Call by value and call by reference

(ii) Storage classes in C

(iii) Hard disk drive

(iv) Organization of a computer

(v) Two dimensional array

(vi) Pointer.



Scanned with CamScanner

Total No. of printed pages = 3

**1 MCA 2 CP&PS**

**2012**

(December)

# COMPUTER PROGRAMMING AND PROBLEM SOLVING

Paper : 102

Full Marks – 75

Pass Marks – 34

Time – Three hours

The figures in the margin indicate full marks for the questions.

Answer any *five* questions.

1. (a) What do you mean by flow chart and algorithm ? Write an algorithm and draw flow chart to obtain the first 5 numbers of a Fibonacci sequence. 8

(b) Draw the block diagram of a computer system. What is pseudo code ? 7

[Turn over

Scanned with CamScanner

2. (a) Write the difference between the following : 2×4=8

(i) High level language and low level language

(ii) Compiler and Interpreter.

(b) Write briefly about Documentation, Debugging, Storage and Computation time estimation. 7

3. (a) What is recursion ? Write a C program to find the factorial of a number using recursion. 7

(b) What do you mean by static and dynamic memory allocation ? Explain dynamic memory allocation using examples in C language. 8

4. (a) What is a file ? Write a C program to count characters, spaces, tabs and new lines in a file. 7

(b) What is a structure ? Write a C program that demonstrates the usage of structure pointer. 8



5. (a) What is sorting and searching ? Write a C program for linear search in an array. 8

(b) Write a C program to sort 10 elements using any sorting algorithm. 7

6. (a) What is an array ? Write a C program to add two matrices. 8

(b) What is pointer ? Write a C program to swap the values of two variables using pointer. 7

7. Write short notes on any *three* : 5×3=15

(i) Storage classes in C

(ii) Function

(iii) Break and Continue Statement

(iv) String

(v) RAM and ROM.



Scanned with CamScanner

*Total number of printed pages–4*

**1 SEM MCA (NCBCS) 2 CPPS**

**2015**

(December)

**COMPUTER APPLICATION**

Paper : 102

***( Computer Programming and Problem Solving )***

*Full Marks : 75*

*Pass Marks : 34*

Time : Three hours

***The figures in the margin indicate full marks for the questions.***

*Answer question no **1** and **any four** from the rest*

1. *(a)* What is a source program? 1×15=15

   *(b)* What is Kernal?

   *(c)* What is additional operator?

   *(d)* Define terniary operator?

*Contd.*

Scanned with CamScanner

(e) What is the significance of % $x$ control specifiers ?

(f) What is the output of printf ("\a") ?

(g) Which statement causes the loop to be terminated ?

(h) What is the difference between 'a' and "a" in C ?

(i) What is enum declaration ?

(j) Differentiate P and *P ?

(k) What is the use of fclose ( ) function ?

(l) Which statement use to define a macro ?

(m) What do you mean computer animation ?

(n) What is the ASCII value of the character # ?

(o) What is meant by text mode ?

2. (a) How does pointer arithmatic help a programmer ? 3

(b) What do you mean searching and hashing ? 3

(c) What is Static and Dynamic memory allocation ? 5



(d) Briefly describe about Operating System ? 4

3. (a) Write down the commonly used library functions in string manipulation header file ? 3

(b) What is preprocessor ? Explain the various preprocessor directives ? 4

(c) Explain various file handling functions available in C ? 5

(d) What is recursion and union ? 3

4. (a) What is primary memory, cache memory and virtual memory ? 6

(b) Write a C programme using pointers to find the biggest of given list of $n$ integers ? 6

(c) What is the difference between compiler and assembler ? 3

5. (a) Briefly describe the storage classes in C ? 5

(b) Write a C programme to reverse a string with the help of user define function ? 6

(c) Write the algorithm and flowchart to calculate compound interest for a given time and principle amount ? 4



Scanned with CamScanner

6. (a) Briefly describe nested for statement and nested if else statement? 6

(b) List the characteristics of First and Third generation computer? 5

(c) Briefly describe about magnetic memory? 4

7. Write short notes on (***any three***) : 5×3=15

(1) Call by value and call by reference.

(2) Debugger and common mistakes.

(3) Describe

int a;
int a [10];
structure variable?
structure array?

(4) Stages of development of language.

(5) Organisation of a computer.

———



Scanned with CamScanner

*Total number of printed pages–4*

**1 SEM MCA (NCBCS) CP & PS 2**

**2016**

(December)

**COMPUTER APPLICATION**

***(Computer Programming and Problem Solving)***

Paper : 102

*Full Marks : 75*

*Pass Marks : 34*

Time : Three hours

***The figures in the margin indicate full marks for the questions.***

*Answer Question No.* ***1*** *and* ***any four*** *from the rest.*

1. *(a)* What is the significance of % *x* control specifier ? 1

   *(b)* What is meant by resolution of a monitor ? 1

   *(c)* What is the ASCII value of the character '—' ? 1

*Contd.*

Scanned with CamScanner

(d) What are the characteristics of 4th generation computers ? 3

(e) What is an object program ? 1

(f) What is a preprocessor ? Explain the various preprocessor directives. 4

(g) What do you mean by structure variable and structure array ? 2

(h) Differentiate $p$ & $*p$. 1

(i) Why do we use continue statement ? 1

2. (a) What do you mean by time and space complexity of an algorithm ? 5

(b) Briefly explain about kernal. 5

(c) What do you mean by function prototype actual argument and formal argument ? Explain with example. 5

3. (a) Explain various file handling function in C. 5



(b) Write a program to perform multiplication of two matrices $A$ and $B$. 7

(c) Explain how a file is opened and closed. 3

4. (a) Write a program to print prime numbers within a range. 6

(b) Write a program to print the fibonacci series. 5

(c) What do you mean by recursion ? Explain. 4

5. (a) What is pointer with arrays ? Write a program using pointer to read an array of integers and print its element in the reverse order. 7

(b) Briefly describe the storage classes in C. 5

(c) What is the difference between contiguous and non-contiguous memory allocation ?



Cont

Scanned with CamScanner

6. (a) Briefly describe 'switch' statement with general syntax. 5

(b) Briefly describe sorting and searching. Write a program to search an element in a given list of element using binary search. 4+6=10

7. Write short notes on : ***(any three)*** 3×5=15

(a) for, while and do-while loop comparision.

(b) Operators in C.

(c) Compiler *vs* interpreter.

(d) Structure in C language.

(e) Call by value and call by reference.



Scanned with CamScanner

*Total number of printed pages–4*

**1 SEM MCA (NCBCS) CP&PS 2**

**2017**

( December )

**COMPUTER APPLICATION**

Paper : 102

***( Computer Programming and Problem Solving )***

*Full Marks : 75*

*Pass Marks : 34*

Time : Three hours

***The figures in the margin indicate full marks for the questions.***

*Answer Question No. **1** and **any four** from the rest.*

1. *(a)* What is the ASCII value of the character "$" ? 1

   *(b)* What is a text editor ? 1

   *(c)* Write down the relation between main memory, auxiliary memory and processor. 3

*Contd.*

Scanned with CamScanner

(d) How many byte occupied in RAM by "Long double"? 1

(e) What is the use of sizeof operator? 2

(f) What do you mean space and time complexity of an algorithm? 4

(g) Differentiate *m* and **m*? 2

(h) Why delay () function is used? 1

2. (a) What are the differences of second and third generation computer? 4

(b) What do you mean memory hierarchy? What are the differences between semiconductor memory and magnetic memory? 6

(c) Write down the preprocessor directives with example. 5

3. (a) Briefly describe the generation of computer languages. 5

(b) Write a C programme to copy the contents of one file to another file. 5



(c) Write short notes: 5

(i) Conditional statement

(ii) Switch statement.

4. (a) What do you mean sorting and searching? Write a C programme to search and element in an array of *n* elements, after sorting the array. 7

(b) Explain with example

(i) for – Loop

(ii) while – Loop

(iii) do–while Loop

(iv) Nested for – Loop. 8

5. (a) Explain the four string handling function in C. Write a C program to test whether a given string is palindrome or not without using string handling function. 8

(b) What is the difference between Structure and Union? Briefly describe structure within structure with a suitable example. 7

6. (a) Explain with example the pointer concept in C. State the advantage of pointer in C. 6



Scanned with CamScanner

(b) What is malloc ( ) and realloc ( ) ? 3

(c) What is debugger ? 3

(d) Define initgraph ( ), closegraph ( ). 3

7. Write short notes on : ***(any three)*** 3×5=15

(a) Flow chart and algorithm.

(b) User defined function and Recursion.

(c) Call by value and call by reference.

(d) Software, Hardware and Kernal.

(e) Conditional and bitwise operator.

---



Scanned with CamScanner

sh
3

r
2

*Total number of printed pages–3*

**1 SEM MCA (NCBCS) 2**

**2018**

( December )

**COMPUTER PROGRAMMING AND PROBLEM SOLVING**

Paper : 102

***( Computer Networks )***

*Full Marks : 75*

*Pass Marks : 34*

Time : Three hours

***The figures in the margin indicate full marks for the questions.***

*Answer question no.* **1** *and* ***any four*** *from the rest.*

1. *(a)* What do you mean by object programme ? 1

   *(b)* What is logical operator ? 2

   *(c)* What is the difference between static memory allocation and dynamic memory allocation ? 3

   *(d)* What do you mean by operator and operand ? 2

*Contd.*

Scanned with CamScanner

(e) What are the advantages of gatche () over scanf () ? 3

(f) What are base address and pointer arithmetic ? 3

(g) What is SVGA ? 1

2. (a) Explain the computer software and its generation. 5

(b) What are relational operators, conditional operators and bitwise operators ? 5

(c) Write a C programe to convert a line in the lower text to upper case. 5

3. (a) What are structure and union ? How does a structure differ from an array ? 5

(b) What is a preprocessor ? Explain various preprocessor directives. 6

(c) What do you mean debugger ? 4

4. (a) Explain, with examples, various file handling functions available in C. 6

(b) Write a C program to find the biggest number from a given list of *n* real values using pointers. 6



(c) Explain switch statement with a suitable example. 3

5. (a) What are the differences between call by value and call by reference ? 5

(b) Explain nested if-else and nested for statement with suitable examples. 7

(c) Explain different data types in C. 3

6. (a) Write a C program to implement matrix multiplication. 7

(b) Briefly explain storage classes in C. 6

(c) Distinguish between actual and formal arguments. 2

7. Write short notes on ***any three*** of the following : 3×5=15

(i) Operating System

(ii) Algorithm and Flow chart

(iii) Dynamic memory

(iv) Conventional language

(v) String handing functions.



Scanned with CamScanner

*Total number of printed pages–3*

**1 SEM MCA (NCBCS) 102**

**2019**

(December)

**COMPUTER APPLICATION**

Paper : MCA-102

***( Computer Programming and Problem Solving )***

*Full Marks : 75*

Time : Three hours

***The figures in the margin indicate full marks for the questions.***

*Answer **any five** questions.*

1. *(a)* What do you mean by assignment, logical and bitwise operators ? 5

   *(b)* What is the time and space complexity of an algorithm ? 5

   *(c)* Describe the generation of a computer. 5

2. *(a)* Distinguish between machine, assembly and high-level language. 5

*Contd.*

Scanned with CamScanner

(b) What are the differences between : 5

(i) Semiconductor memory and Auxiliary memory ?

(ii) Software and Hardware ?

(c) Write down algorithm and draw the flowchart for finding greatest number from given two numbers. 5

3. (a) Briefly describe Nested-for loop with a suitable example. 8

(b) What do you mean by Call by value and Call by reference ? 7

4. (a) What do you mean by sorting and searching ? Implement binary search technique. 7

(b) Explain with example, the various file handling functions available in C. 5

(c) What is actual and formal argument in C ? 3



5. (a) Explain with example, various string handling function in C. 6

(b) What is the difference between Structure and Union ? 2

(c) Write a C program to convert decimal number to binary number. 7

6. (a) What is a preprocessor ? Explain the various preprocessor directives. 6

(b) Write a C programme using pointer to add three numbers. 6

(c) What do you mean by data types in C ? 3

7. Write short notes on : ***(any three)*** 5×3=15

(a) Operating System

(b) Compiler

(c) Computer organization

(d) Bubble sort

(e) Memory hierarchy.



Scanned with CamScanner